

1. Abstract needs tightening and a results preview

The abstract currently repeats the bit-flip explanation from the introduction almost verbatim and doesn't mention any quantitative outcome (e.g., that RTL simulation confirmed correct ACK/NACK generation, or the CRC-32 polynomial used). Recommend condensing the motivation to 2–3 sentences and adding one sentence summarizing the verification result, so a reader can gauge the contribution without reading the full paper.

2. Title promises "error detection coverage analysis" that isn't delivered

The title explicitly claims a coverage analysis, but Section IV only shows a single simulated waveform (Fig. 3) and a synthesis snapshot (Fig. 4) — there's no fault-injection study, coverage metrics (e.g., % of injected bit-errors correctly flagged as NACK), or corner-case testing (burst errors, all-zero/all-one payloads). Either (a) add a small coverage table/plot showing detection rate across injected error patterns, or (b) revise the title to match the actual scope (e.g., drop "coverage analysis").

3. Modulo-2 arithmetic example is disconnected from the actual design parameters

Section II-C works through a manual CRC division using a 4-bit divisor (1011) and a short bit string, which is pedagogically fine, but it doesn't connect back to the paper's actual CRC_POLY (32'h04C11DB7) or the 32-bit MAGIC_RESIDUE described just above it in II-A. Consider either scaling the worked example to reflect the real polynomial (or an explicit note that it's a simplified illustrative example) so readers can trace the logic from the toy example to the implemented hardware.

4. Insufficient methodological detail on the RTL/verification setup

Section III states the project "involves writing synthesizable RTL code (Verilog)" and Section IV states "RTL level verification is completed," but there's no mention of the simulation tool, testbench structure, clock frequency/timing constraints, target FPGA/ASIC library for synthesis (Fig. 4 shows a synthesized netlist with no context), or area/timing/power results. Adding a brief subsection with tool names, test vector generation strategy, and post-synthesis timing/area numbers would substantially strengthen the results section.

5. Figures and language need polish before final submission

Figure 1 (the X25 shift-register schematic) is low-resolution and its caption is missing; Figure 3's waveform is too small to verify signal values described in the text. There are also minor grammatical/typographical issues throughout (e.g., "Pheripheral," "Cyclic Redundacy," "checkswhether," reference [1]'s garbled publisher line, inconsistent duplicate numbering in [2] and [3]). Recommend a full proofreading pass and higher-resolution figure exports before camera-ready submission.

Recommendation: Accept with minor revisions. The core contribution — a configurable, parameterized CRC-32 generator/checker for PCIe LCRC with basic RTL verification — is sound and relevant to the journal's scope; the issues above are addressable without a new experimental campaign.